

Introduction to Game Theory

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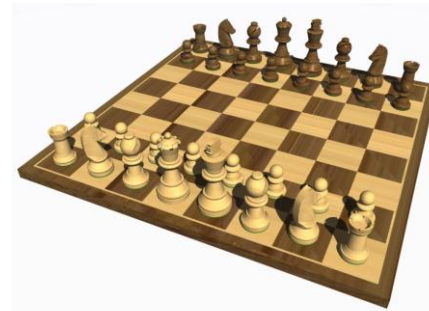
Summer, 2026

Today

- Motivation
- Concepts in game theory
- Examples
- Solution approaches

Motivation

- Life is full of conflict and competition
 - Military battles
 - Political campaigns
 - Advertising and marketing campaigns by competing business firms
 - Auctions
- Decision makers (adversaries) interact with each other
 - The payoff is determined by decisions of all decision makers
- Example: in chess or the game of go
 - I know my best move
 - You know my best move
 - I know you know my best move
 - ...



Motivation: Prisoner's Dilemma



- Two members of a criminal gang are arrested and held separately, unable to communicate (Non cooperative)
- The police lack enough evidence to convict them on the principal charge and can only sentence each to 2 years in prison based on existing evidence
- The police offer each prisoner a deal:
 - If one testifies against the other while the other remains silent, the betrayer goes free, and the silent prisoner receives a 10 year sentence
 - If both prisoners testify against each other, they each receive a reduced sentence of 5 years

Motivation: Prisoner's Dilemma



		Prisoner B	
		Stay silent	Betray
Prisoner A	Stay Silent	Both serve 2 years	Prisoner A serves 10 years Prisoner B goes free
	Betray	Prisoner B serves 10 years Prisoner A goes free	Both serve 5 years

Motivation: Prisoner's Dilemma

		Prisoner B	
		Stay silent	Betray
Prisoner A	Stay silent	-2 , -2	-10 , 0
	Betray	0 , -10	-5 , -5

- Suppose both Prisoners A and B optimize their decisions
- What are their optimal strategies?

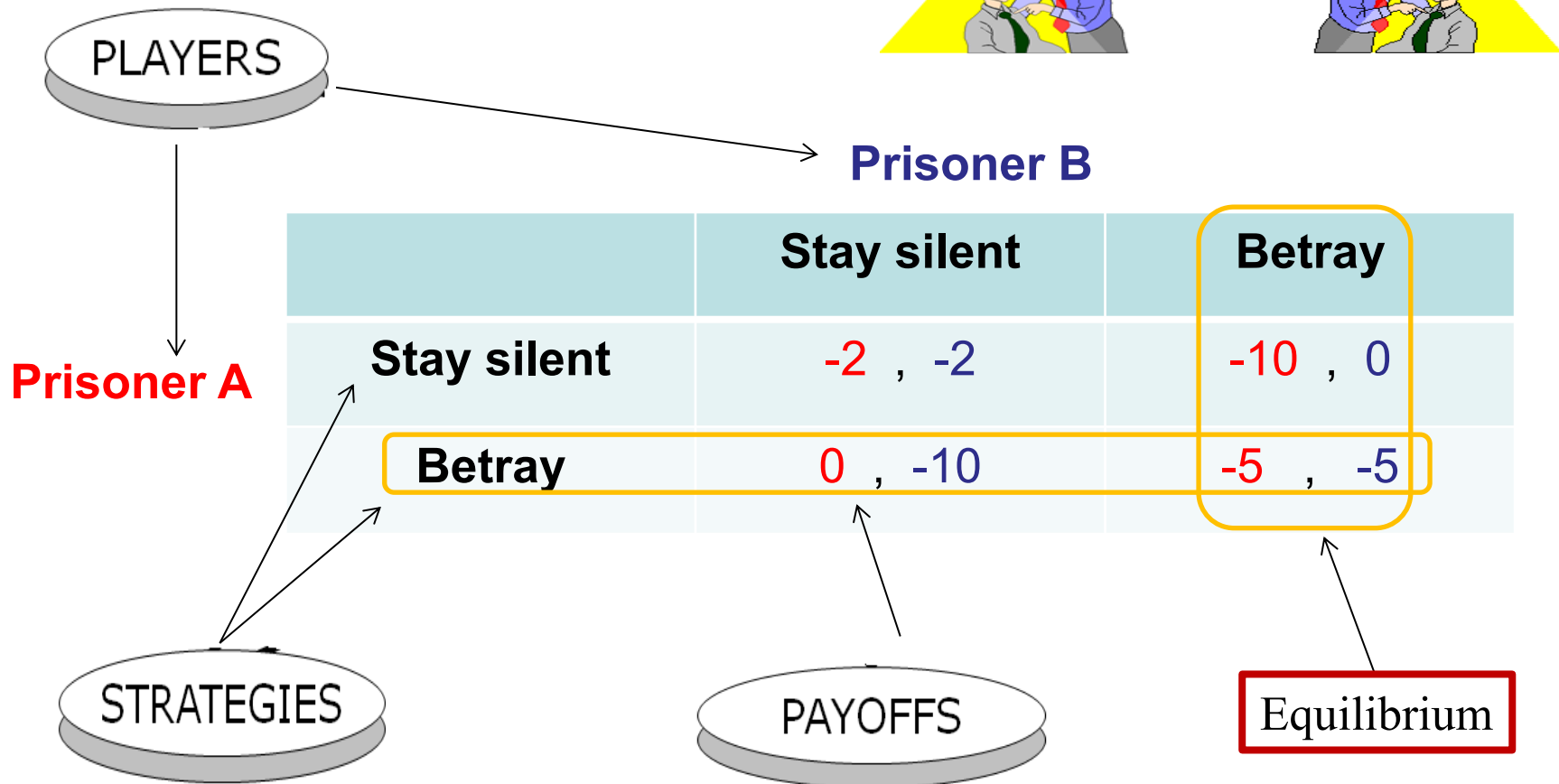
Both Betray!

Game Theory

- Game theory studies strategic decision making
- Mathematically capture the behaviors in strategic situations (games), in which an individuals' success in making choices depends on the choices of others
- The optimal solution to a game is called an equilibrium which consists of decisions of all players. If individuals deviate from the equilibrium, they are worse off
- An important branch of operations research and economics
 - Eight game theorists won Nobel prize



Game Theory



What is a Game

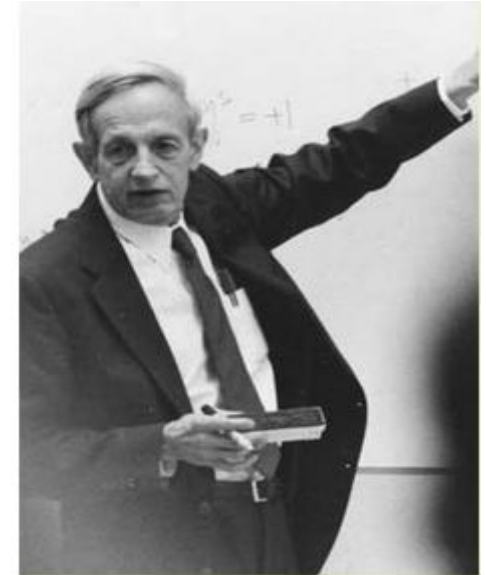
- **Game:** Any set of circumstances that has a result dependent on the actions of two or more decision makers
- **Players:** Strategic decision makers within the context of the game
 - **Rationality & maximization:** Players are rational and will strive to maximize their payoffs/utilities in the game
 - **Self-interested:** A decision maker will maximize one's own payoff in the game (no compassion for the opponent)
 - **Common knowledge:** Each decision maker has perfect knowledge of the game
- **Strategies:** A predetermined rule that specifies completely how one intends to respond to each possible circumstance
- **Payoff table:** It shows the gain for a player that would result from each combination of strategies for the other players

Equilibrium

- A Nash Equilibrium occurs when no player can gain an advantage by unilaterally changing their strategy, assuming all other players keep their strategies unchanged
- In other words:
 - I am making the best possible choice given your actions
 - You are making the best possible choices given my actions
- Essentially, it is a stable state where no single player has an incentive to deviate from their chosen strategy

Nash Equilibrium

- John F. Nash Jr. (1928 – 2015)
 - American mathematician, a pioneering researcher in game theory
- In 1994, he was awarded the Nobel Prize in Economics for his groundbreaking work in game theory



John F. Nash Jr. (1928 – 2015)

Professor S. Lefschetz
Department of Mathematics
Princeton University
Princeton, N. J.

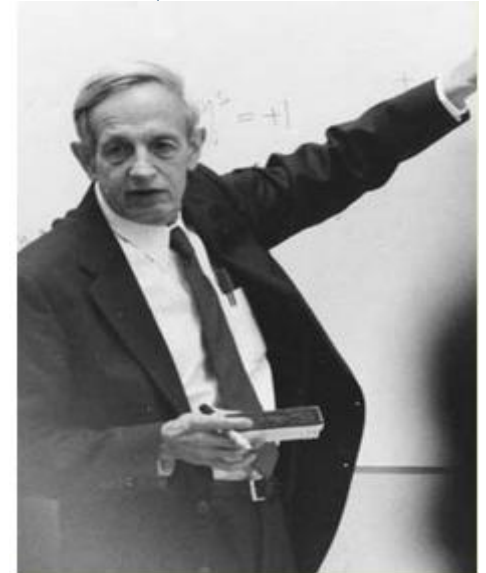
Dear Professor Lefschetz:

This is to recommend Mr. John F. Nash, Jr.
who has applied for entrance to the graduate college
at Princeton.

Mr. Nash is nineteen years old and is
graduating from Carnegie Tech in June. He is a
mathematical genius.

Yours sincerely,

Richard P. Duffin



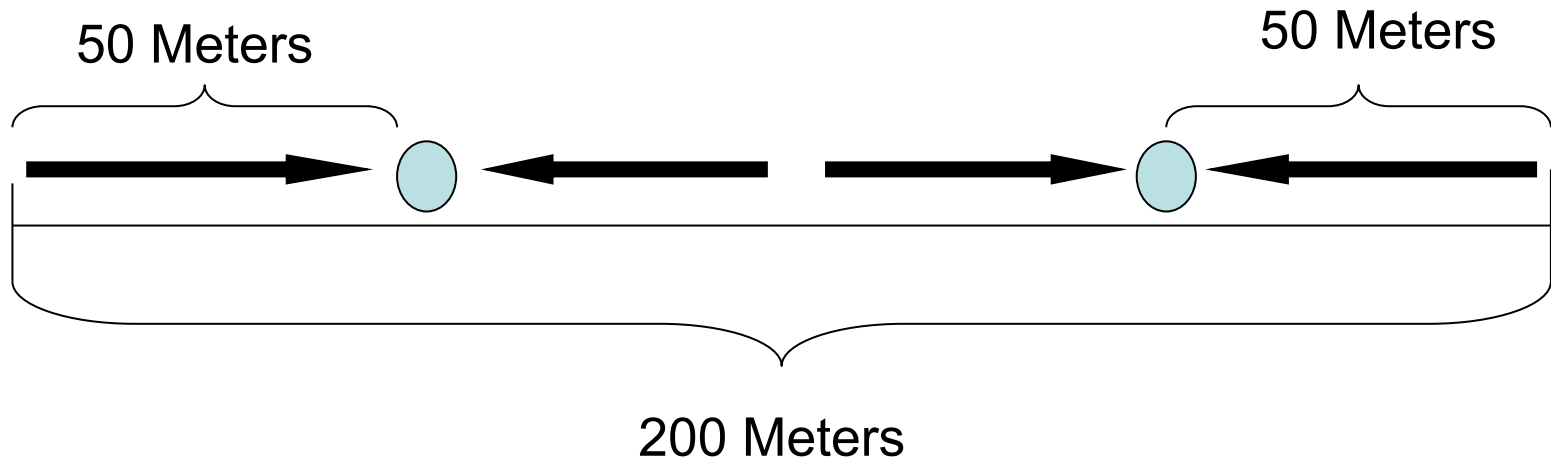
Beach Kiosk Game

- Imagine a beach that is 200 meters long
- There are two vendors licensed to sell soft drinks, snacks, etc. They can set up their kiosks anywhere they want on the beach. Their goal is to attract as many customers as possible
- People are uniformly located along the beach and prefer buying from the nearest kiosk
- **Question: Where will the kiosks be located?**



Beach Kiosk Game

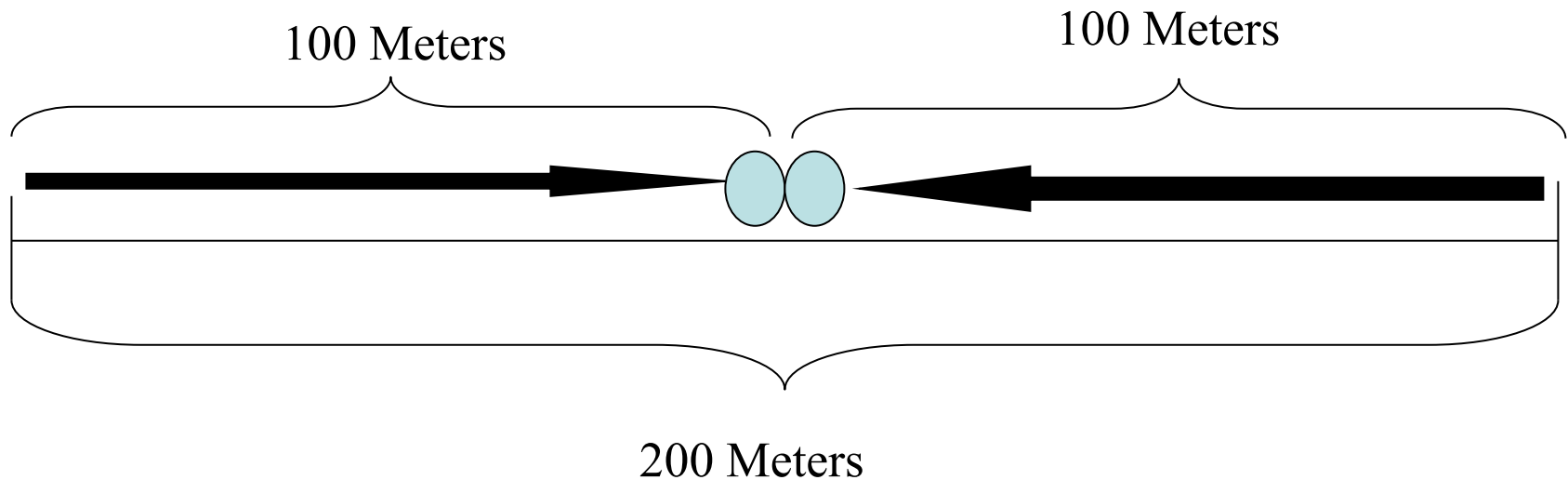
- One choice is to locate a kiosk 50 meters from the end of the beach, and the other 50 meters from the other end
- Each serves half of the beach



- Is this a Nash equilibrium?

Beach Kiosk Game

- In this way, both companies would like to move towards the middle
- In the end, the Nash equilibrium is as follows

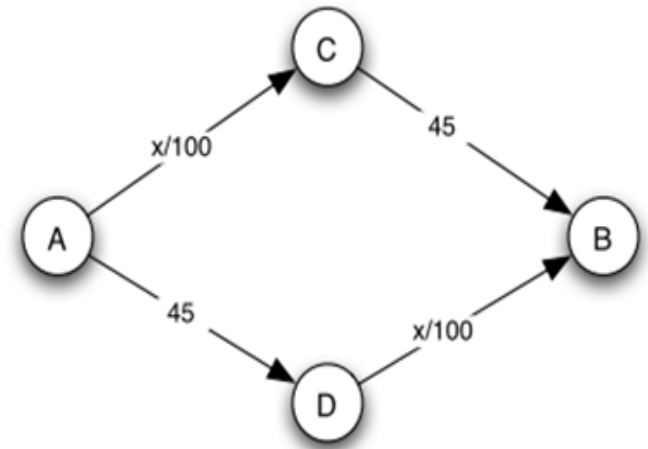


Beach Kiosk Game

- The only stable equilibrium is where both vendors are right next to each other, in the center of the beach
 - Neither can move without losing customers
 - Both the original and the final positions equally split the traffic
- However, at equilibrium, people on the beach have to walk a longer distance
 - Stable, but inefficient market behaviors
- Businesses in competitive markets often cluster: Fast-food chains clustering in city centers rather than spreading evenly

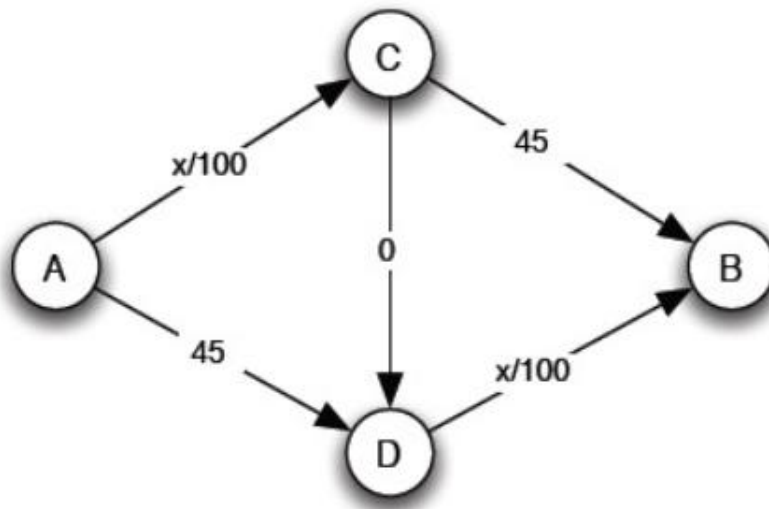
Traffic Analysis

- Traffic flows from source A to sink B
- The roads connecting A to C and D to B are narrow and have limited capacities, which means that travel time depends on the volume of traffic
- Suppose that there are 4000 cars from A to B and everyone aims to minimize the total travel time, what is the Nash equilibrium?



Traffic Analysis

- Suppose that a new road CD is added to the network and suppose that the time for traveling from C to D is 0.
- Will the equilibrium travel time be shorter?

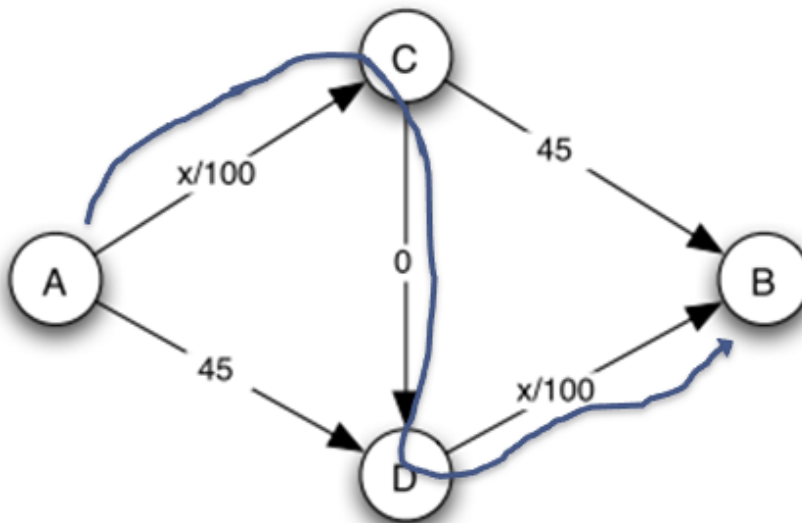


What will happen?



Traffic Analysis

- The equilibrium is that all traffic chooses A to C to D and to B, which requires a total travel time of 80 minutes.

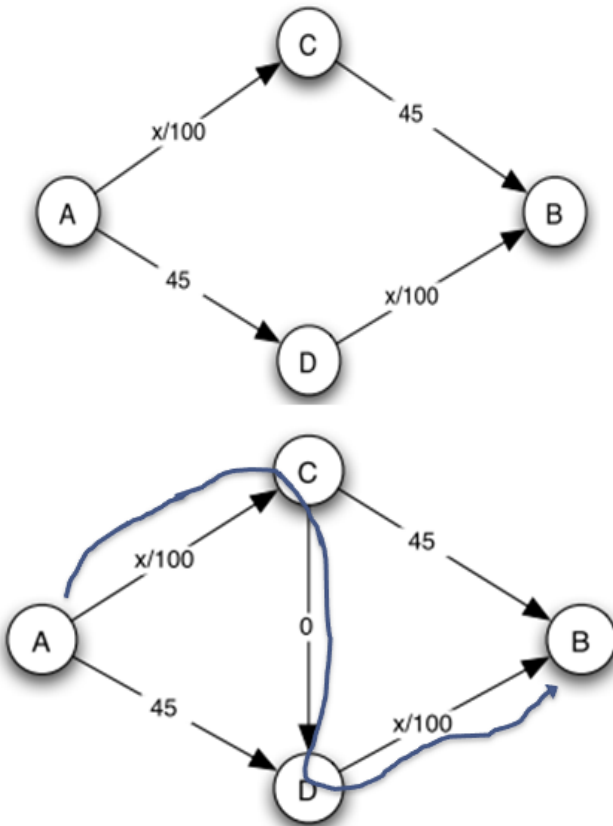


Everyone
picks the
ACDB route.



Traffic Analysis

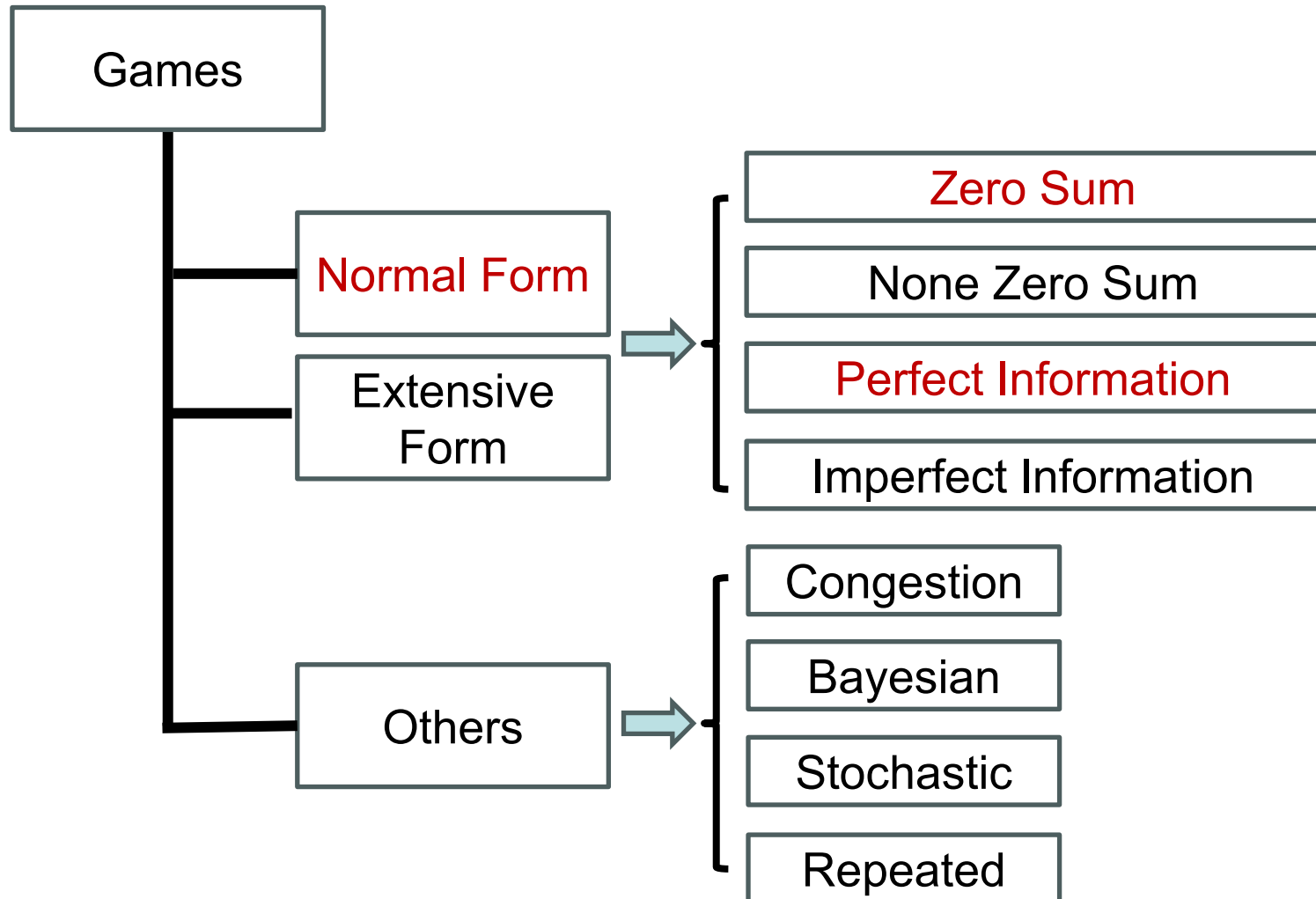
- This phenomenon is named Braess's Paradox: adding resource to a transportation network can sometimes hurt performance as equilibrium.



Braess's Paradox In Reality

- In Stuttgart, Germany, after investments into the road network in 1969, the traffic situation did not improve until a section of newly-built road was closed for traffic again
- In Seoul, South Korea, traffic around the city sped up when the Cheonggye Expressway was removed as part of the Cheonggyecheon restoration project
- In 2009, New York experimented with closures of Broadway at Times Square and Herald Square, which resulted in improved traffic flow and permanent pedestrian plazas

Taxonomy of games



Finding Nash Equilibriums in a Two-Player Game

- Working example:
 - Two politicians are running against each other
 - Both of them want to spend two days campaigning in two key cities, A and B
 - To avoid wasting campaign time, they plan to travel at night and spend either one full day in each city or two full days in just one of the cities
 - However, since arrangements must be made in advance, neither politician will learn his (or her) opponent's campaign schedule until after he has finalized his own
 - This is a zero-sum game, i.e., one player wins whatever the other one loses, so that the sum of their net winnings is zero
- Each politician has three strategies
 - Strategy 1: Spend one day in each city
 - Strategy 2: Spend both days in A
 - Strategy 3: Spend both days in B

Variation 1 of the working example

- Suppose the payoff (the total net votes won in units of 1,000 votes) table for **Player 1** is

Strategy		Player 2		
		S1	S2	S3
Player 1	S1	1	2	4
	S2	1	0	5
	S3	0	1	-1

- A strategy is dominated by a second strategy if the second strategy is always at least as good (and sometimes better) regardless of what the opponent does

Variation 1 of the working example

- Suppose the payoff (the total net votes won in units of 1,000 votes) table for **Player 1** is

Strategy		Player 2		
		S1	S2	S3
Player 1	S1	1	2	4
	S2	1	0	5
	S3	0	1	-1

- Eliminating strategy 3 for Player 1 yields

Strategy		Player 2		
		S1	S2	S3
Player 1	S1	1	2	4
	S2	1	0	5

Variation 1 of the working example

- Eliminating strategy 3 for Player 2 yields

Strategy		Player 2		
		S1	S2	
Player 1	S1	1	2	
	S2	1	0	

- Eliminate strategy 2 for Player 1 yields

Strategy		Player 2		
		S1	S2	
Player 1	S1	1	2	

Variation 1 of the working example

- Eliminating strategy 2 for Player 2 yields

Strategy		Player 2		
		S1		
Player 1	S1	1		

- Finally, we obtain

Value of the game. If it is 0, then the game is a fair game

Strategy		Player 2		
		S1		
Player 1	S1	1		

Back to Prisoner's Dilemma

- We get the equilibrium



Prisoner A

Prisoner B

	Stay silent	Betray
Stay silent	-2 , -2	-10 , 0
Betray	0 , -10	-5 , -5

↑
Nash Equilibrium

Exercise: Simplify the game

		Player 2				
		S1	S2	S3	S4	S5
Player 1	S1	3	2	0	3	0
	S2	5	0	2	5	9
	S3	7	3	9	5	9
	S4	4	6	8	7	5.5
	S5	6	0	8	8	3

Variation 2 of the working example

- Suppose the payoff (the total net votes won in units of 1,000 votes) table for **Player 1** is

Strategy		Player 2			Minimum
		S1	S2	S3	
Player 1	S1	-3	-2	6	-3
	S2	2	0	2	0
	S3	5	-2	-4	-4
Maximum		5	0	6	

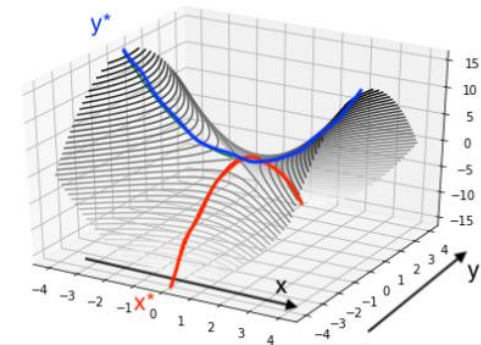
- The minimax criterion is a standard criterion proposed by game theory for selecting a strategy
 - Player 1 should select the strategy whose minimum payoff is largest
 - Player 2 should choose the one whose maximum payoff to player 1 is the smallest

Variation 2 of the working example

- Suppose the payoff (the total net votes won in units of 1,000 votes) table for **Player 1** is

Strategy		Player 2			Minimum
		S1	S2	S3	
Player 1	S1	-3	-2	6	-3
	S2	2	0	2	0
	S3	5	-2	-4	-4
Maximum		5	0	6	

- The minimax criterion is a standard criterion proposed by game theory for selecting a strategy
 - The solution point is a saddle point, because it is both the minimum in its row and the maximum of its column.



Variation 3 of the working example

- Suppose the payoff (the total net votes won in units of 1,000 votes) table for **Player 1** is

Strategy		Player 2			Minimum
		S1	S2	S3	
Player 1	S1	0	-2	2	-2
	S2	5	4	-3	-3
	S3	2	3	-4	-4
Maximum		5	4	2	

- The solution suggested by the minimax criterion is unstable
- The key point is that if the player announces its decision, the opponent can take advantage of it
- So it is necessary to choose among alternative acceptable strategies on some kind of random basis

Variation 3 of the working example

- Suppose the payoff (the total net votes won in units of 1,000 votes) table for **Player 1** is

Strategy			Player 2		
			0	1/2	1/2
			S1	S2	S3
Player 1	1/2	S1	0	-2	2
	1/2	S2	5	4	-3
	0	S3	2	3	-4

- Compared to the previous table where players have pure strategies, the current table shows the expected payoff for Player 1 with mixed strategies (due to the probabilities for each strategy)
- What is the expected payoff in this table?

Variation 3 of the working example

- Suppose the payoff (the total net votes won in units of 1,000 votes) table for **Player 1** is

Strategy			Player 2		
			y_1	y_2	y_3
			S1	S2	S3
Player 1	x_1	S1	0	-2	2
	x_2	S2	5	4	-3
	x_3	S3	2	3	-4

- Compared to the previous table where players have pure strategies, the current table shows the expected payoff for Player 1 with mixed strategies (due to the probabilities for each strategy)
- What is the expected payoff in this table?

- Suppose the payoff (the total net votes won in units of 1,000 votes) table for **Player 1** is

Strategy			Player 2		
			y_1	y_2	y_3
			S1	S2	S3
Player 1	x_1	S1	0	-2	2
	x_2	S2	5	4	-3
	x_3	S3	2	3	-4

- To find the mixed strategies at equilibrium, we need to figure out probabilities for Player 1 and 2
- Minimax theorem: If mixed strategies are allowed, the pair of mixed strategies that is optimal according to the minimax criterion provides a stable solution with equal expected payoff for both players

Variation 3 of the working example: Graphical Solution

- Suppose the payoff (the total net votes won in units of 1,000 votes) table for Player 1 is

Strategy			Player 2		
			y_1	y_2	y_3
			S1	S2	S3
Player 1	x_1	S1	0	-2	2
	x_2	S2	5	4	-3
	x_3	S3	2	3	-4

- The second strategy dominates the third strategy

Strategy			Player 2		
			y_1	y_2	y_3
			S1	S2	S3
Player 1	x_1	S1	0	-2	2
	$1-x_1$	S2	5	4	-3

Variation 3 of the working example: Graphical Solution

- Suppose the payoff (the total net votes won in units of 1,000 votes) table for **Player 1** is

Strategy			Player 2		
			y_1	y_2	y_3
			S1	S2	S3
Player 1	x_1	S1	0	-2	2
	$1-x_1$	S2	5	4	-3

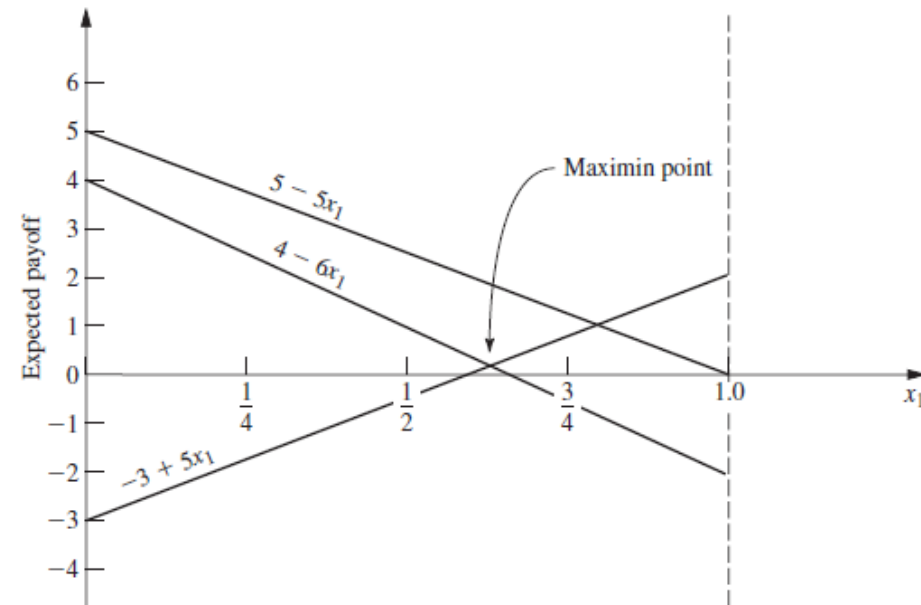
- Expected payoffs for Player 1 if Player 2 use pure strategies

(y_1, y_2, y_3)	Expected payoff
(1,0,0)	$0 \cdot x_1 + 5 \cdot (1-x_1)$
(0,1,0)	$-2 \cdot x_1 + 4 \cdot (1-x_1)$
(0,0,1)	$2 \cdot x_1 - 3 \cdot (1-x_1)$

Variation 3 of the working example: Graphical Solution

- Graphical solution given the expected payoffs

(y_1, y_2, y_3)	Expected payoff
(1,0,0)	$0 \cdot x_1 + 5 \cdot (1-x_1)$
(0,1,0)	$-2 \cdot x_1 + 4 \cdot (1-x_1)$
(0,0,1)	$2 \cdot x_1 - 3 \cdot (1-x_1)$



- The optimal mixed strategies for Player 1 is determined by expected payoff in the second and third rows
 - $(x_1, x_2) = (7/11, 4/11)$
 - Expected payoff is $2/11$

Variation 3 of the working example: Graphical Solution

- To find the optimal mixed strategies for Player 2, we need to use the minimax theorem
- The payoff table becomes

Strategy			Player 2		
			y_1	y_2	y_3
			S1	S2	S3
Player 1	7/11	S1	0	-2	2
	4/11	S2	5	4	-3

- The optimal mixed strategies for Player 1 implies that

$$0y_1^* - 2y_2^* + 2y_3^* = 2/11$$

$$5y_1^* + 4y_2^* - 3y_3^* = 2/11$$

- In addition, we have $y_1^* = 0$, and $y_1^* + y_2^* + y_3^* = 1$
- The optimal mixed strategies for Player 2 is $(0, 5/11, 6/11)$

Variation 3 of the working example: LP Solution

- Suppose the payoff (the total net votes won in units of 1,000 votes) table for Player 1 is

Strategy			Player 2		
			y ₁	y ₂	y ₃
			S1	S2	S3
Player 1	x ₁	S1	0	-2	2
	x ₂	S2	5	4	-3

- Two-player zero-sum games with mixed strategies can be solved by transforming the problem to a linear programming problem

$$\begin{aligned}
 &\max_{x_1, x_2, v_x} && v_x \\
 &s.t. && 5x_2 \geq v_x \\
 &&& -2x_1 + 4x_2 \geq v_x \\
 &&& 2x_1 - 3x_2 \geq v_x \\
 &&& x_1 + x_2 = 1 \\
 &&& x_1, x_2 \geq 0
 \end{aligned}$$

$$\begin{aligned}
 &\min_{y_2, y_3, v_y} && v_y \\
 &s.t. && -2y_2 + 2y_3 \leq v_y \\
 &&& 5y_1 + 4y_2 - 3y_3 \leq v_y \\
 &&& y_1 + y_2 + y_3 = 1 \\
 &&& y_1, y_2, y_3 \geq 0
 \end{aligned}$$

Market Competition

- Duopoly market:
 - Two firms competes in a market where price function is known as $p = a - bq$, where q is the total quantity of the product.
 - Their products are assumed to be the same with the same unit production cost c .
 - They are free to choose production quantity with no capacity limit.
 - Two firms simultaneously choose their production quantity

Cournot Model

- All firms produce a homogeneous product, i.e., there is no product differentiation
- Firms do not cooperate, i.e., there is no collusion
- Firms have market power, i.e., each firm's output decision affects the good's price
- The number of firms is fixed
- Firms compete in quantities rather than prices
- The firms are economically rational and act strategically, usually seeking to maximize profit given their competitors' decisions

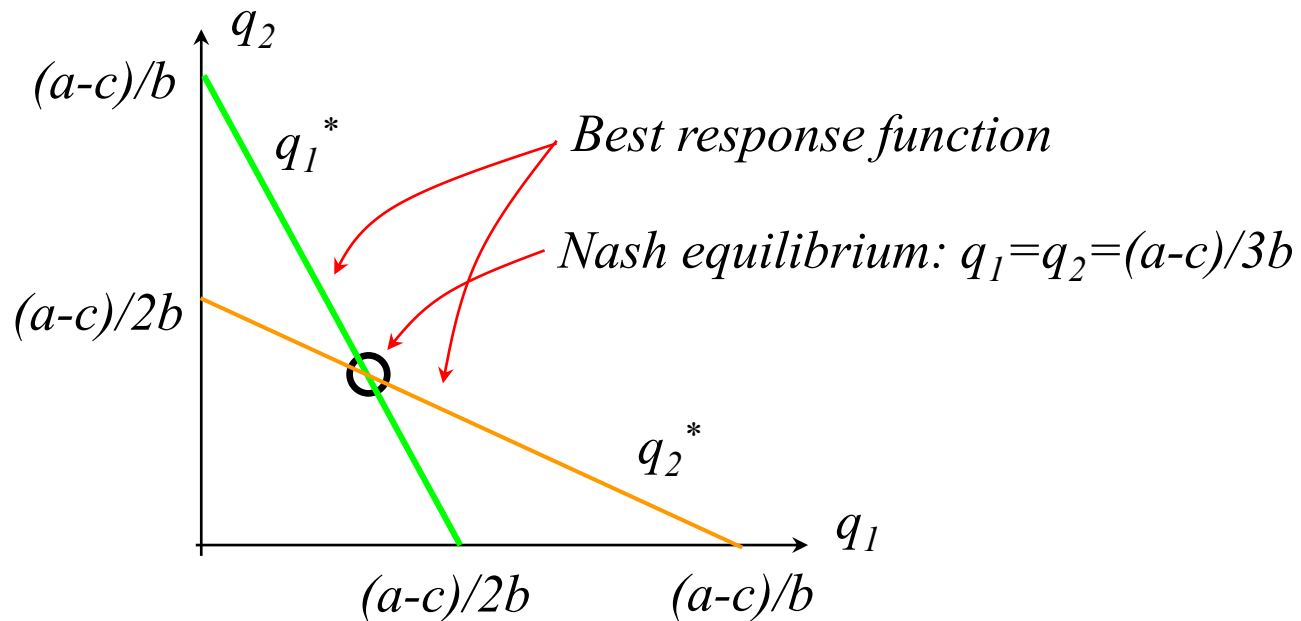
Cournot Model

- Two firms produce the same product in quantities of q_1 and q_2 , respectively
 - Market clearing price $p = a - b (q_1 + q_2)$
- Profit for firm 1
 - $\pi_1 = (p - c) q_1 = [a - b (q_1 + q_2) - c] q_1$
- Objective: choose q_1 to maximize profit
 - $\max [a - b (q_1 + q_2) - c] q_1$
 - we get $q_1^* = (a - c - bq_2)/(2b)$
- Similarly, $q_2^* = (a - c - bq_1)/(2b)$

Cournot Model

- Given the competitor's decision, the best response function:

- $q_1^* = (a - c - bq_2)/(2b)$
 $q_2^* = (a - c - bq_1)/(2b)$



- Nash equilibrium: $q_1^* = q_2^* = (a-c)/3b$, $\pi_1^* = \pi_2^* = (a-c)^2/9b$

Practice Problem

■ Duopoly market:

- Two firms competes in a market where price function is known as $p = 14 - 2q$, where q is the total quantity of the product.
- The production cost of firm 1 is 2, and the cost of firm 2 is 4.
- They are free to choose production quantity with no capacity limit.
- Two firms simultaneously choose their production quantity
- Solution: $q_1^* = 7/3$; $q_2^* = 4/3$